mcu

File: mcu.kicad_sch

CN7 / CN10: DAC, ADC, I2C, clocks

dac

File: dac.kicad_sch

level shift 1.65 V -> 2.5 V mid, V_cm reference

adc

File: adc.kicad_sch

8 x series R, BAT54S clamp, anti-alias C

power

File: power.kicad_sch

jack, LDO +5VA, TCA9548A

tiles

File: tiles.kicad_sch

JT1 -> tile 1, JADC <- last tile

designed, ERC-checked, not built, not measured

dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /

File: carrier.kicad_sch

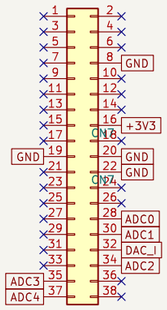
Title: Carrier board for NUCLEO-G474RE + tiles (M3a)

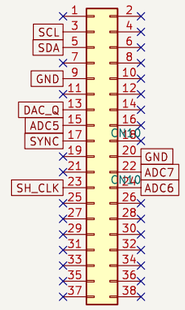
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 1/6

NUCLEO-G474RE Morpho headers (female 2x19, CN7 left / CN10 right). Unused pins no-connect. Pin map verified against UM2505 Rev 4 Table 16 (see gen_carrier.py header).

CN7: 16=3V3, 28=PA0 ADC1_IN1, 30=PA1 ADC1_IN2, 32=PA4 DAC1_OUT1, 34=PB0 ADC1_IN15, 35=PC2 ADC2_IN8, 37=PC3 ADC2_IN9

CN10: 3=PB8 I2C1_SCL (AF4), 5=PB9 I2C1_SDA (AF4), 13=PA6 DAC2_OUT1, 15=PA7 ADC2_IN4, 17=PB6 GPIO (SYNC), 22=PB2 ADC2_IN12, 23=PA8 TIM1_CH1 (AF6), 24=PB1 ADC1_IN12





dlnn phase 2: designed, ERC-checked, not built, not measured

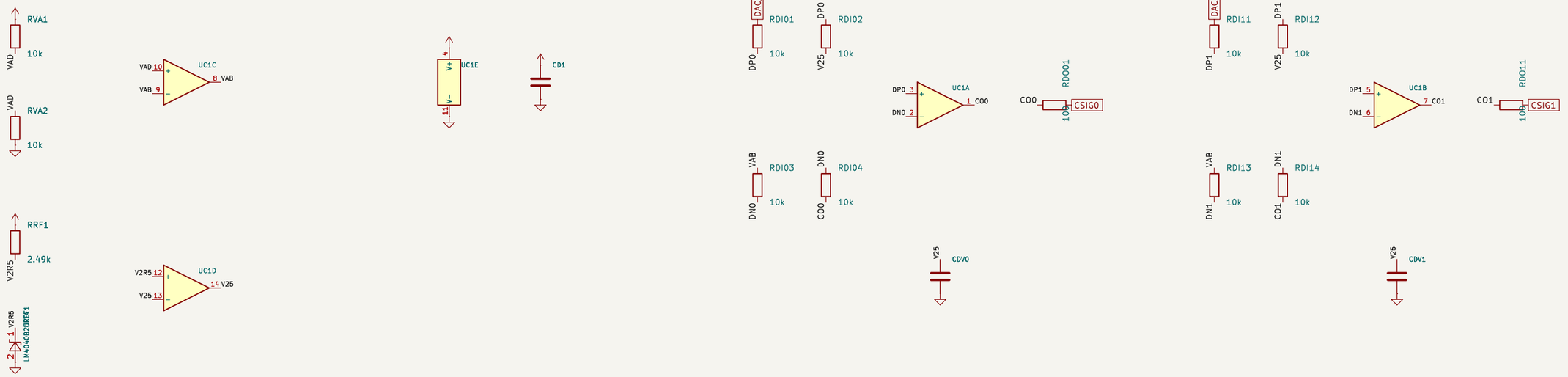
Sheet: /mcu/
File: mcu.kicad_sch

Title: Nucleo Morpho headers

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 2/6

DAC I/Q conditioner: V_out = V_dac + 2.5 V − 1.65 V (unity difference amp, 4 x 10k), I/Q centred on 2.5 V for the tile input RC (1k + 330 pF, on the layer−1 tile). Unit 3: 3V3/2 follower, unit 4: V_cm buffer.

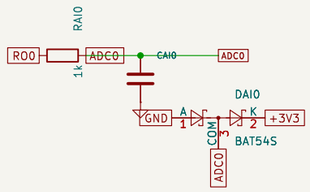
RDO: 100 ohm series build−out into the ribbon

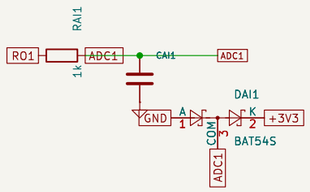


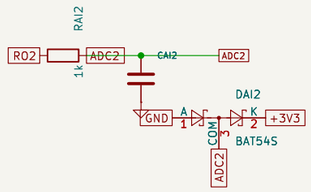
dlnn phase 2: designed, ERC−checked, not built, not measured

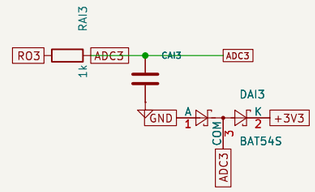
Sheet: /dac/		
File: dac.kicad_sch		
Title: DAC I/Q conditioner		
Size: A3	Date: 2026−11−03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 3/6

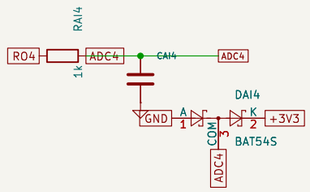
ADC front end: last tile OUT0..7 (ribbon JADC) -> 1k series, BAT54S clamp to +3V3/GND, 330 pF anti-alias (482 kHz) -> Nucleo ADC1/2 inputs ADC0..7

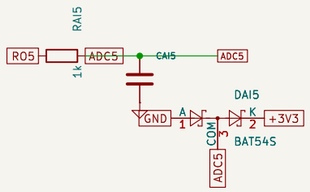


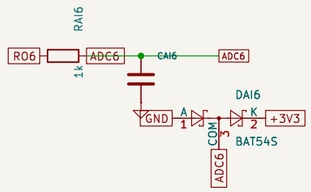


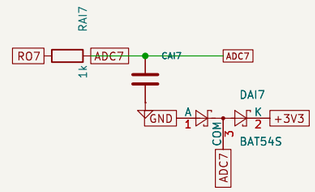












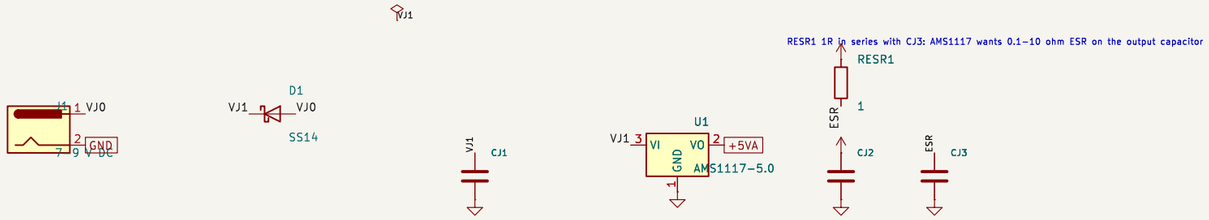
dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /adc/
File: adc.kicad_sch

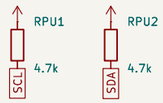
Title: ADC front end

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 4/6

Power: 7–9 V barrel jack -> series Schottky -> AMS1117-5.0 (JLC C6187, Basic, SOT-223, Vin max 15 V, dropout 1.3 V) -> +5VA. MP20051DQ (C134275) dropped: JLC page lists 5.5 V max supply and an adjustable 0.8–5 V output, unusable from a 7–9 V jack. A 7 V supply is preferred (dissipation (Vin−5 V) x 0.22 A).



I2C host pull-ups 4.7k (SCL, SDA, 3.3 V) for the shared bus; per-tile TCA9548A (address 0x70+ID) sit on the tiles



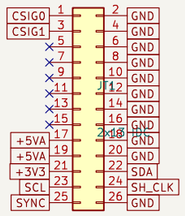
dlnn phase 2: designed, ERC-checked, not built, not measured

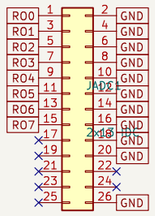
Sheet: /power/
File: power.kicad_sch

Title: Power and I2C mux

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 5/6

Ribbon headers: JT1 -> tile 1 J_IN (I/Q on SIG0/SIG1, power, shared I2C, S/H clock, sync); tile 1 J_OUT -> tile 2 J_IN -> tile 3 J_OUT -> JADC (8 signals). One shared I2C bus; each tile has its own TCA9548A (ID jumpers).





dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /tiles/
File: tiles.kicad_sch

Title: Tile ribbon headers

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 6/6